Tomographic phase microscopy

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Supplementary figures and text:

Supplementary Figure 1 Phase projection geometry.

Supplementary Figure 2 Bright field image and index tomogram of *C. elegans*.

Supplementary Figure 3 Tomogram images from a 30 μm thick suspension of polystyrene beads.

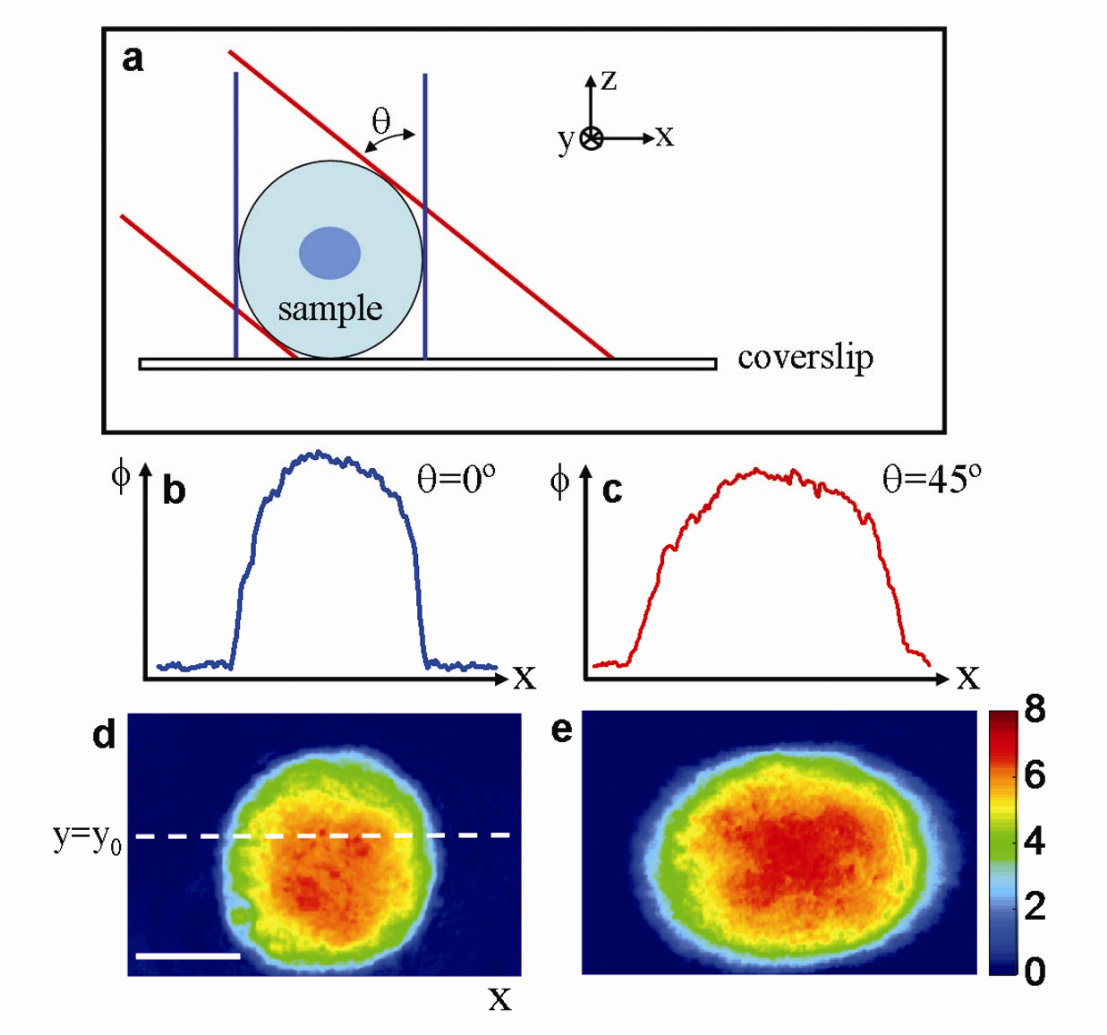
Supplementary Figure 4 Tomogram images from an 80 μm thick suspension of polystyrene beads.

Supplementary Methods

Supplementary Software Matlab code for 3-D reconstruction.

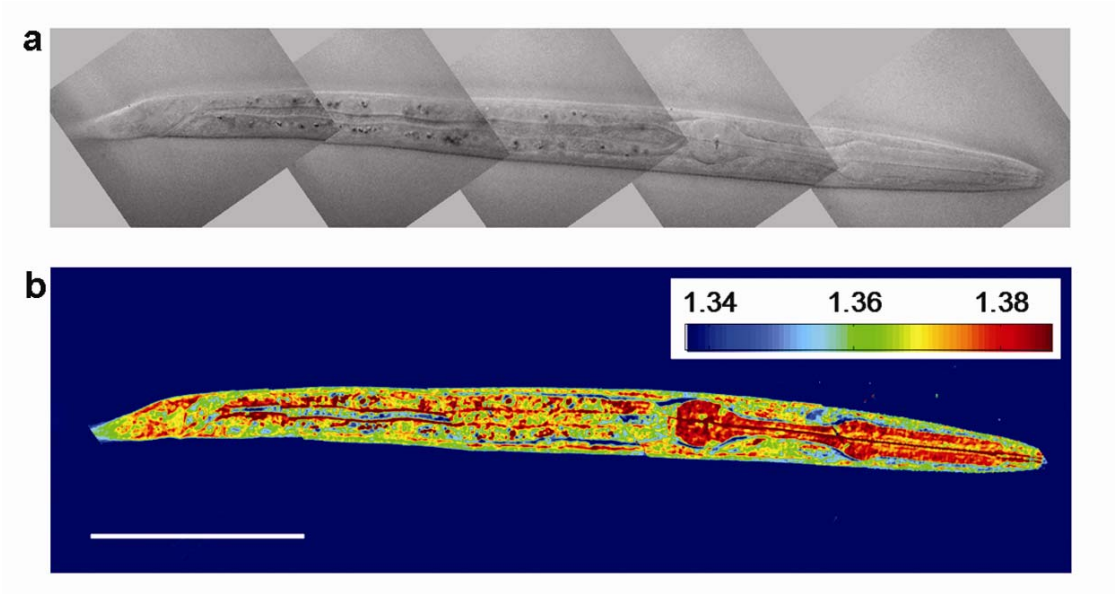
Note: Supplementary Video 1 is available on the Nature Methods website.

Supplementary Figure 1. Phase projection geometry



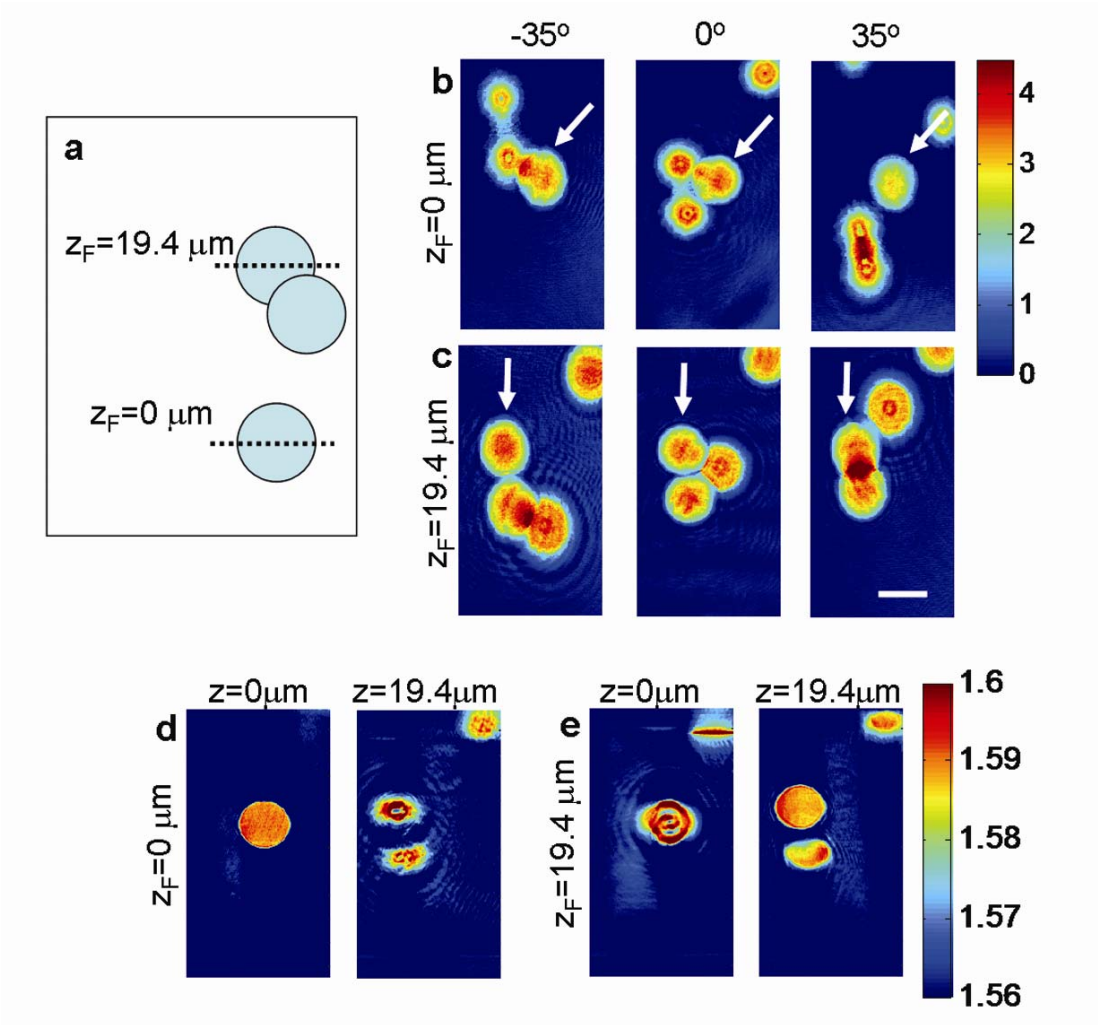
(a) Illumination geometry and projection on x-z plane. Blue lines represent the illumination at zero degrees and red lines at angle θ. (b) and (c) Projection phase line profiles of a HeLa cell at θ = 0 and 45 degrees, respectively, at fixed $y = y_0$. (d) and (e) Projection phase images corresponding to (b) and (c), respectively. The color bar indicates the phase in radian. Scale bar, 10μm.

Supplementary Figure 2. Bright field image and index tomogram of C. *elegans.*



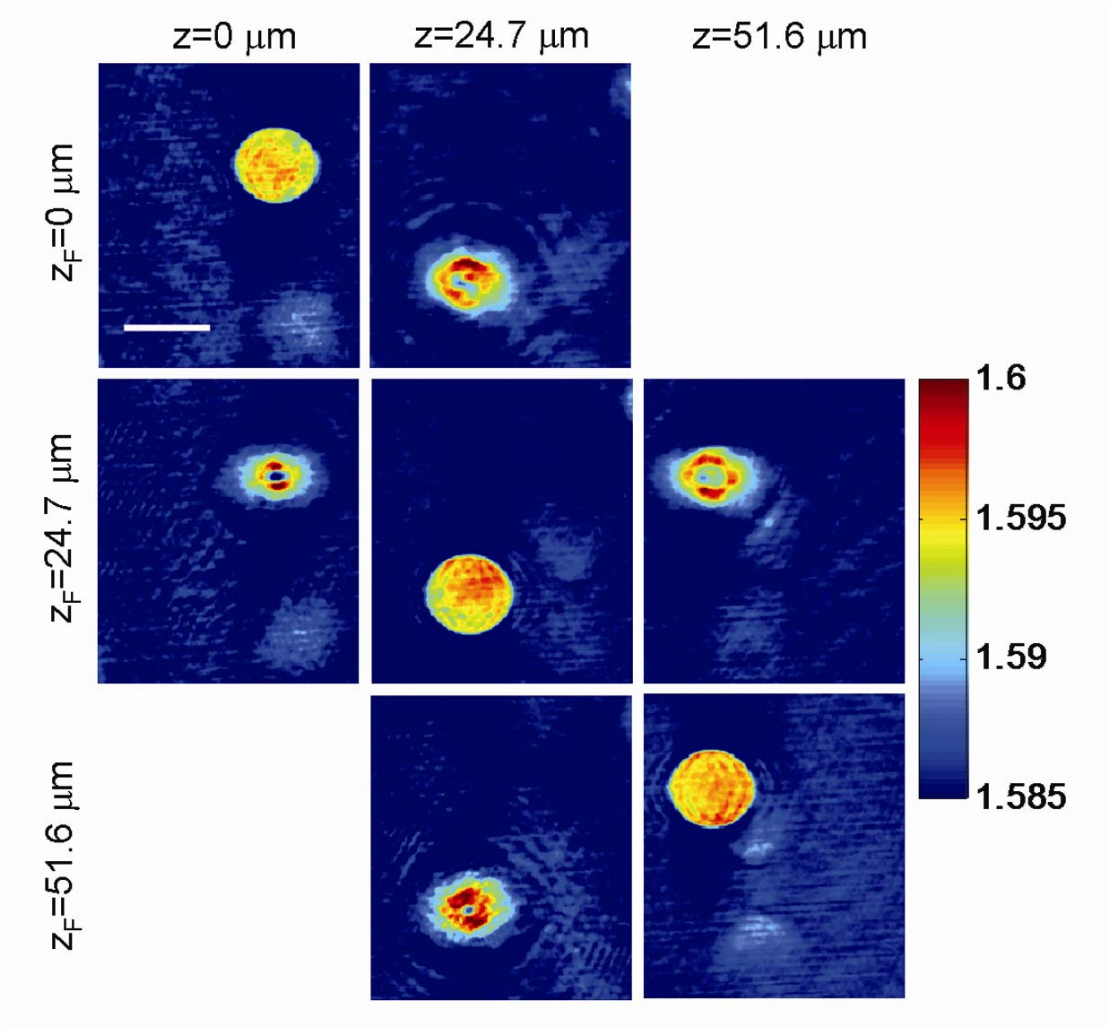
(a) Mosaic of bright field images of a juvenile C. *elegans* nematode. (b) Mosaic of x-y slices of index tomograms through center of worm (same as **Figure 3d** in main text). The color bar indicates the refractive index at $\lambda = 633$ nm. Scale bar, 50 μm.

Supplementary Figure 3. Tomogram images from a 30 μm thick suspension of polystyrene beads



Projection phase images and refractive index tomograms of polystyrene beads in suspension. (a) Sample geometry. (b), (c) Phase images at objective focus z_F = 0 and 19.4 μm, and illumination angles -35, 0 and 35 degrees. The color bar indicates the phase in radians. Scale bar, 10 μm. (d), (e) Tomograms at heights z = 0 and 19.4 μm at objective focus z_F = 0 and 19.4 μm, respectively. The color bar indicates the refractive index at λ = 633 nm.
See **Supplementary Methods** for details.

Supplementary Figure 4. Tomogram images from an 80 μm thick suspension of polystyrene beads



Tomographic images of polystyrene beads in suspension at heights z = 0 μm, 24.7 μm and 51.6 μm and objective focus z_F = 0 μm, 24.7 μm and 51.6 μm. The color bar indicates refractive index at λ = 633 nm. Scale bar, 10 μm. See **Supplementary Methods** for details.

Supplementary methods

Bright field and fluorescence imaging

The tomographic phase microscope is also designed to accommodate bright field and fluorescence imaging. For bright field imaging, a white light emitting diode (LED) placed between the scanning mirror and a condenser lens serves as an illuminator, and images are recorded by a CCD camera (Photometric CoolSnap HQ) along a separate optical path (not shown in **Fig. 1**). Bright field images provide a form of optical sectioning, due to the extremely short depth of field (<1 μm) provided by NA 1.4 illumination and collection. For fluorescence imaging, a standard filter cube (Olympus) with appropriate filters is placed under the objective lens, and fluorescence excitation is provided by either a mercury arc lamp (Olympus) or a collimated blue LED (Lumileds). For single cells, widefield fluorescence imaging using DAPI or SYTO (Invitrogen) nucleic acid stains is used to identify nuclear boundaries.

Laser interferometer phase measurement

In the interferometer sample arm, the beam is incident on a tilting mirror controlled by a galvanometer (HS-15, Nutfield technology) (**Fig. 1**). A lens (L1, f = 250mm) is used to focus the beam at the back focal plane of an oil-immersion condenser lens (Nikon 1.4NA), which recollimates the beam to a diameter of approximately 600 μm. The distances from tilting mirror to the lens (L1) and from the lens (L1) to the back focal plane of the condenser lens are set equal to the focal length of the lens (L1) such that the tilting mirror is conjugate to the sample plane.

Samples are prepared in chambers composed of two glass coverslips separated by a plastic spacer ring and partially sealed with adhesive. Light transmitted through the sample is collected by an infinity-corrected, oil-immersion objective lens (Olympus UPLSAPO 100XO, 1.4 NA). A tube lens (L2, f = 200 mm) focuses an image of the sample onto the camera plane with magnification M=110.

In the reference arm, the laser beam passes through two acousto-optic modulators (AOMs) (Isomet) driven at frequencies $\omega_1 = 110.1250$ MHz and $\omega_2 = 110.0000$ MHz, respectively, using a custom-built digitally synthesized RF driver. Irises select the +1 and -1 order beams, respectively, such that the total reference beam frequency shift is 1250 Hz. After passing through the AOMs, the reference beam is spatial filtered and enlarged by a beam expander.

A beamsplitter recombines the sample and reference laser beams, forming an interference image at the camera plane. The focus and tilt of sample and reference beams are adjusted to minimize the difference between the two wavefronts.

For each phase image, a high speed CMOS camera (Photron 1024-PCI, 17μm pixel size) records 4 images separated by 200 μs, exactly one-quarter the reciprocal of the heterodyne frequency. In this way, four interference patterns I_1, I_2, I_3, I_4 are recorded in which the sample-reference phase shift between consecutive images differs by $\pi/2$. Phase images are then obtained by applying phase shifting interferometry using the four-

bucket algorithm[1] $\phi(x,y)=\arg(((I_4-I_2) + i(I_3-I_1))$. 2π-phase ambiguities are resolved by phase unwrapping using Goldstein's algorithm[2]. The fast acquisition time between frames reduces the effects of external noise. Exposure times are typically ~20 μs. By stepping the galvanometer mirror, 81 phase images are recorded for sample illumination angles θ = -60 to +60 degrees in steps of 1.5 degrees.

The phase projection geometry is shown in **Supplementary Figure 1**, illustrated by data from a single HeLa cell in a culture medium (DMEM with 10% fetal bovine serum). The phase image of the cell for θ = 0 degree is shown in **Supplementary Figure 1d**. For nonzero θ the image displays a background fringe pattern due to the tilt between sample and reference beams, with fringe spacing $d = \lambda M/(n \sin \theta)$, with M = 110 the magnification and n the index of the sample medium (image next to the camera in **Figure 1** of the main text). The corresponding phase image shows a 2π jump at every fringe. After unwrapping the phase, the phase image appears as a nearly linear phase ramp with the phase profile of the cell almost indistinguishable.

The phase image at an illumination angle of 45° is shown in **Supplementary Figure 1e** after phase unwrapping and subtraction of the background phase ramp via a least-square linear fit to a portion of the field of view with no sample. Compared to the zero degree angle case (**Supplementary Fig. 1b**), the line profile of the phase image is elongated at nonzero illumination (**Supplementary Fig. 1c**), because the phase measurements are performed in an image plane parallel to the sample substrate, regardless of incident beam direction (**Supplementary Fig. 1a**).

A similar angle-dependent set of phase images is obtained with no sample present, and the resulting set of background phase images is subtracted from the sample phase images to eliminate residual fixed-pattern phase noise due to optical aberrations and imperfect alignment.

Iterative constraint algorithm

The limitation of projection angles to $|\theta| < 60$ degrees poses a problem of missing information. There have been many computed tomography studies addressing limited angle problems[3-5]. To reduce the effect of the missing projections, we applied an iteration based constraint method[5]. In this technique, the reconstruction is first performed by filling missing-angle projections with values of zero. The resulting reconstructed image, which represents the difference of the refractive index relative to that of the surrounding medium, is constrained to contain only non-negative values and set to zero outside some boundary chosen well outside the margins of the cell. Next, a θ-dependent projection of this reconstruction is calculated and constrained to equal experimentally measured projections over the range of measured angles θ. The process is repeated 10-15 times to ensure convergence.

Absolute index calibration

Since all phase measurements in this experiment are measured relative to other points in the field of view, the tomographic data from the algorithm gives the refractive index relative to that of the medium. The absolute index was calculated by adding the

relative index to the index of refraction of the culture medium, found to be 1.337 using a different interferometric method[6].

Imaging thick samples

The measurement of 3-D refractive index will allow quantitative characterization of sample-induced aberrations. Such aberrations become progressively more severe for thicker tissues, although recent work has shown that biological structures ~30 μm thick can induce significant optical aberrations[7]. We therefore explored methods for imaging samples much thicker than single cell layers.

Our reconstruction algorithm approximates the phase of the sample field as the integral of refractive index along a straight line in the direction of beam propagation. We refer to this as the projection approximation; it is also known as the eikonal or ray approximation. The projection approximation places constraints not only on the index variations of the sample but also its thickness. For plane wave illumination of a typical cell, the projection approximation is accurate to depths of roughly 15 microns. To address this limitation, we have explored the use of focusing at multiple planes to extend the range of tomographic imaging.

We prepared test samples composed of 10 μm polystyrene beads (Polysciences) suspended in UV-curable optical adhesive (Norland or Dymax) and sandwiched between two glass coverslips (**Supplementary Fig. 3a**). Projection phase images at three different angles of illumination (-35°, 0° and 35°) for two different positions of the objective focus (z_F = 0 μm and z_F = 19.4 μm) are shown in **Supplementary Figure 3b-c**. In

Supplementary Figure 3c, the in-focus bead (white arrow) is clearly resolved and satisfies the projection approximation while the out-of-focus beads exhibit ring patterns due to diffraction, demonstrating the breakdown of the projection approximation. Note that for both focus positions, in-focus beads (white arrows) are stationary with respect to change in illumination angle, while out-of-focus beads are not. This angle-dependent position information is the basis of depth discrimination in our tomographic reconstruction algorithm; the in-focus beads are separated in height from the defocused beads although all beads are present in the phase images.

In **Supplementary Figure 3d**, x-y slices of the resulting tomogram at the centers of an in-focus bead (z = 0 μm) and defocused bead (z = 19.4 μm) are shown at an objective focus z_F = 0 μm. The same slices for z_F = 19.4 μm are shown in **Supplementary Figure 3e**. The in-focus beads**,** which satisfy the projection approximation in the phase images, are clearly resolved in the tomograms, while out-of-focus beads are blurred and exhibit ring pattern artifacts. Similar results are obtained in test samples up to 80 μm thick. In **Supplementary Figure 4**, three focal depths are used to measure tomograms centered at three beads, with a 51 μm separation between the highest and lowest. The signal-to-noise ratio is somewhat lower in this data, due to the smaller refractive index contrast between beads and adhesive.

These results suggest a simple scheme for measuring samples of extended thickness, in which the objective focus is automatically scanned over intervals of 15μm (or less) to cover the sample depth and obtain a set of tomograms at each step of the focus.

By combining in-focus slices in series, a mosaic tomogram covering the entire sample can then be created. The maximum thickness of samples is then limited not by the projection approximation but by other factors, such as sample absorption, light scattering and the objective working distance.

Data collection and analysis

Experiment control, data acquisition, data processing and image display were done using MATLAB software (Mathworks Inc.). Data analysis code is given in **Supplementary Software 1**.

References

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Supplementary Software 1

Matlab code for 3-D reconstruction

```
%%% 1. 3D reconstruction using matlab's iradon command.
%%% 2. Non-negative constraint algorithm is applied.
%%% 3. Input data file format: tif
%%% 4. Contents of data:
%%%   In this example, 4 images have been recorded for each of 120 illumination angles
%%%   The sample data is saved as a sequence starting with image_0001.tif and
%%%   ending with image_0480.tif. An equal number of images at the same set of angles are recorded
%%%   as a background, and numbered image_0481.tif through image_0960.tif


%% Initial condition
clear;

datapath='c:\data'; %location of data files
cd(datapath);

Mag = 110;     % magnification of the microscope
res = 17/Mag;   % resolution (in microns) per pixel. In our case pixel size is 17micron
lambda = 0.633 % wavelength (in microns) of light source. In our case, HeNe Laser
index_medium = 1.337; %refractive index of medium.

aa=imread('image_0241.tif'); % Import one arbitrary image to determine the size of image in pixels
figure(501),imagesc(aa);    % Plot the imported image
[xsize ysize] = size(aa)    % Image size is determined. x-axis:parallel to the rotation. yaxis:orthogonal to the rotation
xtick = (1:xsize)*res;      % x axis tick label is created based on the size of the image
ztick=xtick;               % z axis tick label is same as x
ytick =(1:ysize)*res ;      % y axis tick label

tot_step = 120;            % total number of angle steps
ini_frame = 1;             % Initial step of angle to be included in the inverse Radon transform
fin_frame = 120;           % Final step of angle to be included in the inverse Radon transform

%% initialize variables for storing phase images
%%% Phase image will be saved to a variable f_projec(angle step, x, y)
%%% Typically the size of image is too large to be called in one variable.
%%% Split the data into two variables:
%%% f_projec(angle step, x, 1<= y <=ysize/2), s_projec(angle_step, x, ysize/2+1 <= y <ysize)
   ss=0;
   yi1 = floor(ysize/2)*ss+1;      %%% setting cropping area of y from 1 to ysize/2
   yf1 = floor(ysize/2+ysize/2*ss);
   ysize_s = yf1-yi1+1;
   f_projec = zeros(tot_step, xsize, ysize_s);
```

```
    ss=1;
    yi2 = floor(ysize/2)*ss+1;      %%%setting cropping area of y from ysize/2+1 to ysize
    yf2 = floor(ysize/2+ysize/2*ss);
    ysize_s = yf2-yi2+1;
    s_projec = zeros(tot_step, xsize, ysize_s);


%% Phase calculation loop using Phase Shifting Interferometry (PSI) algorithm
    for jj = 1:tot_step % calculate one angle step at a time

        for mm=0:1 % m=0 for background, m=1 for sample

            %%%% load 4 successive images of background and sample at a
            %%%% given angle
            kk = 4*(jj+tot_step*(1-mm)-1)+1;
            fname1= strcat('signal_c001s000',num2str(1000000+kk),'.tif');
            fname2= strcat('signal_c001s000',num2str(1000000+kk+1),'.tif');
            fname3= strcat('signal_c001s000',num2str(1000000+kk+2),'.tif');
            fname4= strcat('signal_c001s000',num2str(1000000+kk+3),'.tif');

            img1 = zeros(xsize, ysize);           % initialize variables
            img2 = img1; img3 = img1; img4 = img1;
            img1 = double(imread(fname1));         % load image to variables
            img2 = double(imread(fname2));
            img3 = double(imread(fname3));
            img4 = double(imread(fname4));

            %%% Quantitative phase calculation
                cphi = (img1-img3)+sqrt(-1)*(img4-img2); %PSI algorithm. cphi = amplitude* exp(i*phase).
                i_projec1 = unwrap2(angle(cphi(:,:)));   %Apply a 2D unwrap algorithm

            %%% eliminate the slope in phase originated from the angle
            %%% difference between sample and reference beam
                if mm==0 % for background phase image
                    endphase(jj) = squeeze(i_projec1(xsize,(ysize-2))-i_projec1(1,(ysize-2))); %obtain the amplitude of the slope
                    inc1 = (i_projec1(round(xsize/2),yini:yfin)-i_projec1(1,yini:yfin))/(round(xsize/2)-1);
                    inc2 = (i_projec1(xsize,yini:yfin)-i_projec1(round(xsize/2)+1,yini:yfin))/(round(xsize/2)-1);
                    inc = mean((inc1+inc2)/2); % calculate the inclination of the slope
                end

                for ll =1:ysize    %subtract the slope
                     projec1(:,ll) = i_projec1(:,ll)-(i_projec1(1,1)+inc*(1:xsize)');
                end

                if mm==1          %store the calculated sample phase image to projec_s
                    projec_s = projec1;
                elseif mm==0       % store the calculated background phase image to projec_b
```

```
                projec_b = projec1;
            end
        end

            projec = projec_s - projec_b; %subtract background phase from sample phase
            figure(501), imagesc(projec), title(strcat('y=',num2str(jj))); %Plot resulting phase image at a given angle
            f_projec(jj,:,:) = projec(:, yi1:yf1); %Store the first part in y axis to f_projec
            s_projec(jj,:,:) = projec(:, yi2:yf2); %Store the second part in y axis to s_projec

    end

        save('phase1.mat','f_projec','endphase','xsize','ysize','ysize_s','tot_step'); %Save all the results to a file
        save('phase2.mat','s_projec','endphase','xsize','ysize','ysize_s','tot_step');
        clear f_projec;
        clear s_projec;

%% Iradon loop with constraint algorithm%%%%

for uu=1:2 %load f_projec first and s_projec later

    if uu==1              %%%load f_projec
        clear projec;
        load('phase1.mat');
        projec = f_projec;
        clear f_projec;
    elseif uu==2           %%%load s_projec
        clear projec;
        load('phase2.mat');
        projec = s_projec;
        clear s_projec;
    end

    %%% angle calibration based on amplitude of slope stored in 'endphase'
    mu_x = endphase/(2*pi)/(17*xsize);      % spatial frequency mu_x at the camera plane in 1/micron
    camera_ang = asin(mu_x/(1/lambda));     % angle at the camera plane in radian
    sample_ang = asin(Mag*sin(camera_ang)/index_medium)*180/pi;
           %angle(in degree) at the sample plane based on the relation n_m*sin(theta_m) = M*sin(theta_camera)
    %%%%%
```

```
end
for ii=1:net_step
   theta_full(ii+ni)=theta(ii);
end
for ii = 1:nf
   theta_full(ii+ni+net_step)= theta(end)-0.25*ii;
end
figure(101),plot(theta_full)

%%%%%%%%%%%%%%%% Initial 3D reconstruction %%%%%%%%%%%%%%%%%%%%%%%%
wsize = xsize;
y_ini = 1;
y_fin = ysize_s;
yysize = y_fin-y_ini+1;
Tomog = zeros(wsize, yysize, wsize);  %set a variable to store the resulting tomogram

for jj = y_ini:y_fin                %load phase image at a given y
   projec1 = squeeze(projec(:,:,jj));  % store original data to projec1
   projec2 = zeros(tot_step, xsize);   % Initialize the projected phase as projec2
   projec2 = projec1;                 % Set projec2 same as projec1
   for ii = 1:tot_step                % Apply cosine factor to projec original image to the normal plane to the beam
                                         propagation direction
      for kk = 1:xsize
         px = round((kk-xsize/2)*cos(sample_angle(ii)*pi/180))+round(xsize/2); %Choose resampling pixel basen on cosine
                                                                                relation
         projec2(ii,px)= projec1(ii,kk);
      end
   end


   %% Apply the inverse Radon transform to get (x,z) out of (theta,x)
   RadonTrans = zeros(tot_step,xsize);
   RadonTrans = medfilt2(projec2',[3 3]);
   RR = RadonTrans(1:1+xsize-1,ini_frame:fin_frame);
   I1 = iradon(RR,theta,'nearest','han',xsize); %Inverse Radon transform with sampled angle
   figure(13), imagesc(I1), xlabel('x'), ylabel('z');

%%%%%% apply constraint algorithm: Using I1 as an initial estimate
      for mm=1:10 % iteration after applying constraint
         if mm == 1  %load initial estimate I1
            II =I1;
         elseif mm >1; %load previously iterated result c_III
            II = c_III ;
         end

         c_II=II;
         for ii=1:xsize        %apply non-negative constraint. Index within the cell is higher than that of the medium
            for kk=1:xsize
```

```
                if c_II(ii,kk)<0
                    c_II(ii,kk)=0;
                end
            end
        end

        %%%% Get a full angle projection of constrained tomogram
        [c_RR,xp] = radon(c_II,theta_full); %Apply Radon transform using theta_full
        [xsize2 ysize2] = size(c_RR)
        xini = floor((xsize2 - xsize)/2+1);
        c_RR1 = c_RR(xini+1:xini+xsize,:);  %Resample the projection to match the size with original projection phase

        %%% Subtitue original projection data
        c_RR2 = zeros(xsize2, ysize2);
        c_RR2 = c_RR1;

            for ii=1:net_step
                c_RR2(:,ii+ni) = RR(:,ii); %Substitute original projection RR within sampled angle range
            end
            figure(16), imagesc(c_RR2), xlabel('theta'), ylabel('x'), colorbar;

        %%% 3D reconstruction with substituted projection
        c_III = iradon(c_RR2,theta_full,'nearest','han',xsize);
        figure(14), imagesc(c_III),xlabel('x'), ylabel('z'),title(strcat('iteration',num2str(mm)));

    end
    %%%%%% end of applying constraint

    CTomog(:,jj-y_ini+1,:) = c_III; %Store the final iterated result
    figure(15), imagesc(c_III), title(num2str(jj));
end

if uu==1
    save('CTomog1.mat','CTomog'); % save the first part of tomogram to a file
    clear CTomog;
    clear projec;
elseif uu==2
    save('CTomog2.mat','CTomog'); % save the second part of tomogram to a file
    clear CTomog;
    clear projec;
end
end
```